



Structured Programming Language

CSE 1111

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Assessment

- Attendance : 5%
- Assignments : 5%
- Class Tests : 20%
- Mid Term : 30%
- Final : 40%

Grading Policy

A (Plain)	: 90-100	C+ (Plus)	: 70-73
A- (Minus)	: 86-89	C (Plain)	: 66-69
B+ (Plus)	: 82-85	C- (Minus)	: 62-65
B (Plain)	: 78-81	D+ (Plus)	: 58-61
B- (Minus)	: 74-77	D (Plain)	: 55-57

Course Outline: Summary

1. Topic 1: Review

- Algorithm, Flow-chart, Pseudocode, Variable, Logic

2. Topic 2: Decision (If/Else-if/Else)

3. Topic 3: Decision (Switch)

4. Topic 4: Loop (For/While/Do-while)

5. Topic 5: Array (1D and 2D)

6. Topic 6: String

7. Topic 7: Nested Loop

8. Topic 8: Function

9. Topic 9: Recursion

10. Topic 10: Structure

11. Topic 11: Pointer

12. Topic 12: File

Resources

1. Programming in C (??? Edition) Stephen G. Kochan
2. Teach Yourself C (3rd Edition or later) Herbert Schildt
3. Programming in ANSI C.(5th Edition or later) E. Balagurusamy
4. URI ONLINE JUDGE
<https://www.urionlinejudge.com.br/judge/en/login>
5. Online Resources:
 - GeeksforGeeks
 - Tutorialspoint

Why Program?

Computer - Programmable machine designed to follow instructions **Instructed by programs.**

Program - A set of instructions that a computer follows to perform a task. Commonly referred to as software (e.g., word, power point).

Programmer - person who writes programs. Without programmers, no programs; Without programs, a computer does nothing

Computer System

A computer system consists of similar hardware devices and software components

– Hardware

: Refers to physical components that a computer is made of.

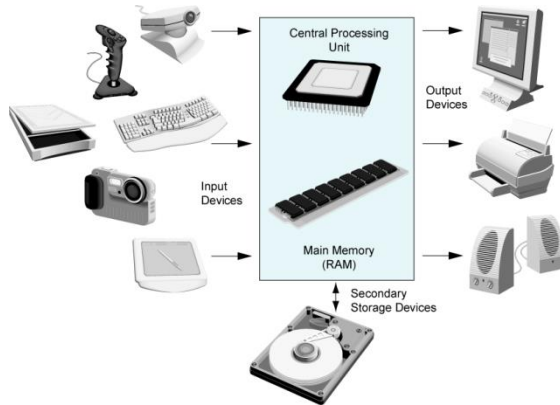


Figure : Organization of a computer system

- Central Processing Unit (CPU)
- Main Memory
- Secondary Memory / Storage
- Input Devices
- Output Devices

– Software

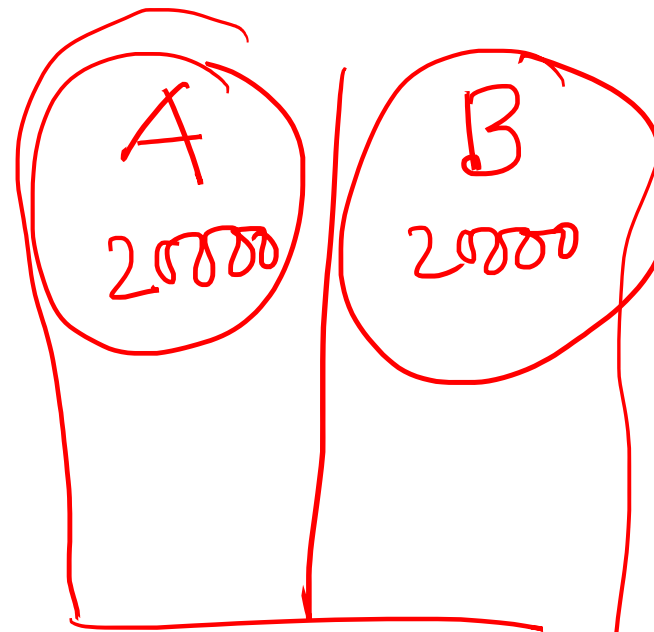
: Makes the physical components to function and perform a task.

$$(3+4) = 7$$

time = X
memory space = *

$$(3800 \times 4000) \% 8 =$$

$$4000^{2000} / \log(3000)$$



① Resource

② Error

Human computation

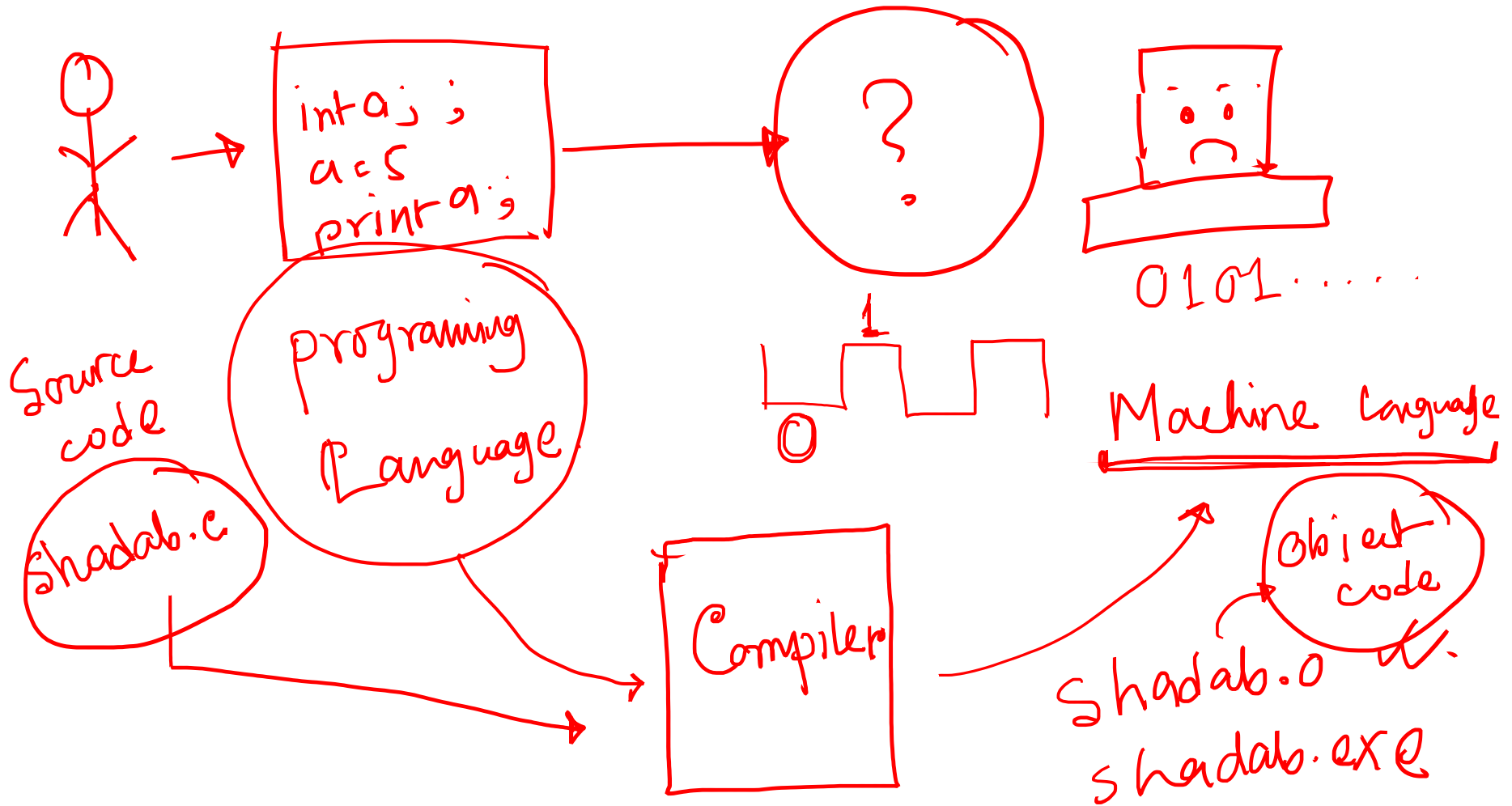
→ automated

FM 3+
5%

Weekly 40hr
5%

UIU Graduate
2%

Weekly
2 hr mis
- 5%



001:

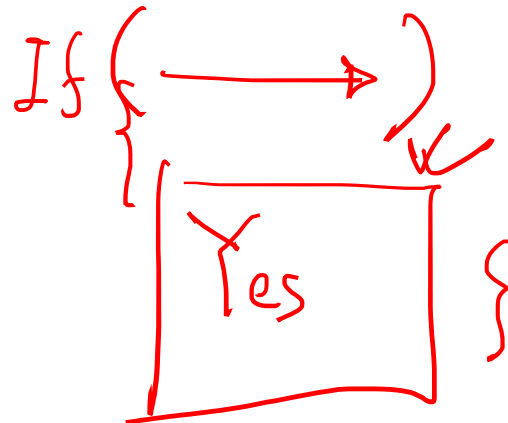
002: CMP A, B

003: JMPNE LABELA

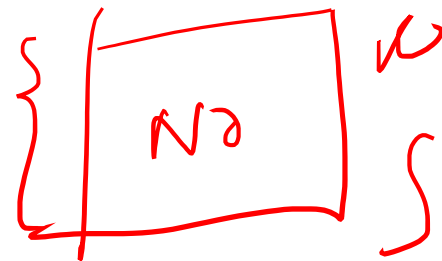
004:

Label A:

~~If else fn~~



else



Central Processing Unit (CPU)

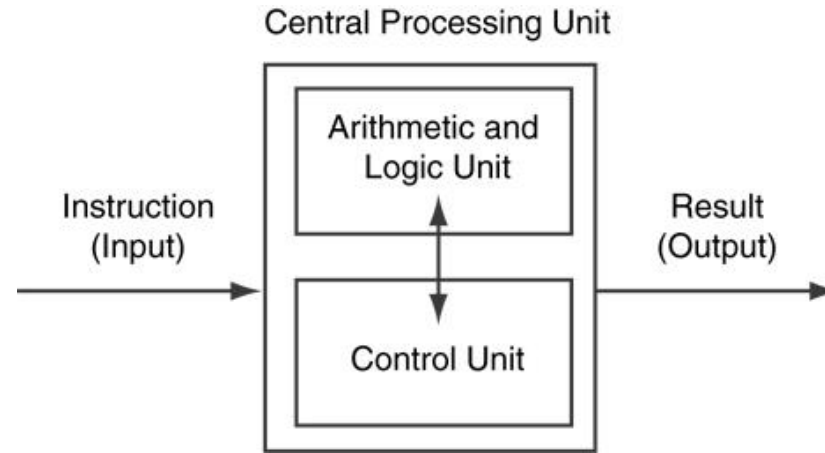
The CPU (Brain of a computer) is the part of a computer that actually runs programs.

- Fetch Instructions
- Follow instructions
- Produce some results

Comprised of:

- Control Unit
 - Retrieves and decodes program instructions
 - Coordinates activities of all other parts of computer
- Arithmetic & Logic Unit
 - It is designed to perform mathematical operations

CPU Organization



When a computer is running a program, the CPU is engaged in a process known formally as the **fetch/decode/execute cycle**.

Fetch : CPU's control unit **fetches the next instruction from main memory**.

Decode: The instruction is encoded in the form of numbers. The **control unit decodes it and generates an electrical signal**

Execute: The **signal is routed to the appropriate component** of the computer to perform an operation.

Main Memory

Random Access Memory (RAM) - Computer's work area.

This is where the computer **stores program and data** while the program is running.

It is volatile. Main memory is erased when program terminates or computer is turned off.

A computer's memory is **divided into tiny storage locations known as bytes**. Each byte is divided into eight smaller storage locations known as **bits**.

bit : smallest piece of memory.

Has values 0 (off, false) or 1 (on, true)

byte: 8 consecutive bits. Bytes have addresses.

Main Memory

Addresses - Each byte in memory is identified by a **unique number known as an address**.

Addresses are ordered from **lowest to highest**.

0	1	2	3	4	5	6	7	8	9
10	11	12	13	14	15	16 149	17	18	19
20	21	22	23 72	24	25	26	27	28	29

The number 149 is stored in the byte with the address 16, and the number 72 is stored at address 23.

Software

Categories of software:

System software: programs that manage the computer hardware and the programs that run on them.

Examples: operating systems, utility programs, software development tools

Application software: programs that provide services to the user.

Examples : word processing, games, programs to solve specific problems



Programs and Programming Languages

To solve a problem using a computer, **one must express the solution** to the problem in terms of the instructions of the particular computer.

A computer *program* is just a collection of the instructions necessary to solve a specific problem. The **approach or method** that is used to solve the problem is known as an ***algorithm***. An ***algorithm*** is **a set of well-defined steps**.

A **Programming language** is a special language **used to write computer programs**.

Machine Language

Algorithms are not ready to be executed on the computer.

The **computer only executes *machine language*** instructions

Machine language instructions are **binary numbers**, e.g.,
1011010000000101

Rather than writing programs in machine language, programmers use *programming languages*.

Types of languages:

Low-level: used for communication with computer hardware directly. Often written in binary machine code (0's/1's) directly. Example: Assembly

High-level: closer to human

Example: C/Java/Python

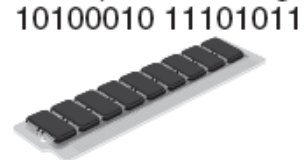
A special computer program

(**compiler**) translates the statements of the program developed in the higher-level language into a form that the computer can understand language.

High level (Easily read by humans)



Low level (machine language)



Structured Programming

Structured language: uses structured constructs of selection (**if/then/else**) and repetition (**while and for**).

Suggested by Corrado Bohm and Guiseppe Jacopini that any computer program can be written with just three structures: **decisions**, **sequences**, and **loops**.

Non-structured language:

- A program in a non-structured language uses unstructured jumps to labels (**Goto**) or instruction addresses.
- The lines are usually numbered or may have labels: this allows the flow of execution to **jump to any line in the program**.
- **Example:** JOSS, FOCAL, TELCOMP, Assembly Languages, early versions of BASIC, Fortran, COBOL



Algorithm

An **algorithm** is

- a step by step sequence of instructions
- to solve a specific problem
- in a finite amount of time

Example 1 (Algorithm)

Find the sum of two numbers and display the result on the screen.

Step 1: Start

Step 2: Input A

Step 3: Input B

Step 4: Calculate $C = A + B$

Step 5: Display C

Step 6: Stop

Corresponding Program

Find the sum of two numbers.

```
# include <stdio.h>
```

```
int main() {
```

```
    int a, b, c;
```

```
    printf("Enter two numbers to add\n");
```

```
    scanf("%d%d", &a, &b);
```

```
    c = a + b;
```

```
    printf("Sum of the numbers = %d\n", c);
```

```
    return 0;
```

```
}
```

Example 2 (Algorithm)

Subtract two numbers and display the result on the screen.

Step 1: Start

Step 2: Input X

Step 3: Input Y

Step 4: Calculate $Z = X - Y$

Step 5: Display Z

Step 6: Stop

Flowchart

Algorithm

Pseudocode

PL code impl

✓✓

Start

input a, b, c

$sum = a + b + c$

print sum

end

Flowchart

Corresponding Program

Subtract two numbers.

```
# include <stdio.h>
```

```
int main() {
```

```
    int x, y, z;
```

```
    printf("Enter two numbers\n");
```

```
    scanf("%d%d", &x, &y);
```

```
    z = x - y;
```

```
    printf("Sum of the numbers = %d\n", z);
```

```
    return 0;
```

```
}
```

What algorithms do you use in everyday life?

Calling a friend on the telephone

Going to the university

Making a cup of tea

Multiply two numbers

Divide two numbers

Summary

All computer systems consist of similar hardware devices and software components

Hardware: Refers to physical components

- Central Processing Unit (CPU)

- Main Memory

- Secondary Memory / Storage

- Input Devices

- Output Devices

Software : Makes the physical components to function

- System Software

- Application Software

Algorithm: A step by step sequence of instructions